NCBA PAYBILL-LEVEL PUSH NOTIFICATIONS

SERVICE GUIDE

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Introduction

About this Guide

This guide provides technical information about integrating and configuring the NCBA Paybill Level Push Notification Service.

Push Notification APIs, both JSON and XML formats are covered in this document for below types of Lipa Na M-Pesa on offer at NCBA bank.

i. Buygoods
ii. Paybill
iii. NCBA Till via 880100

Intended Audience

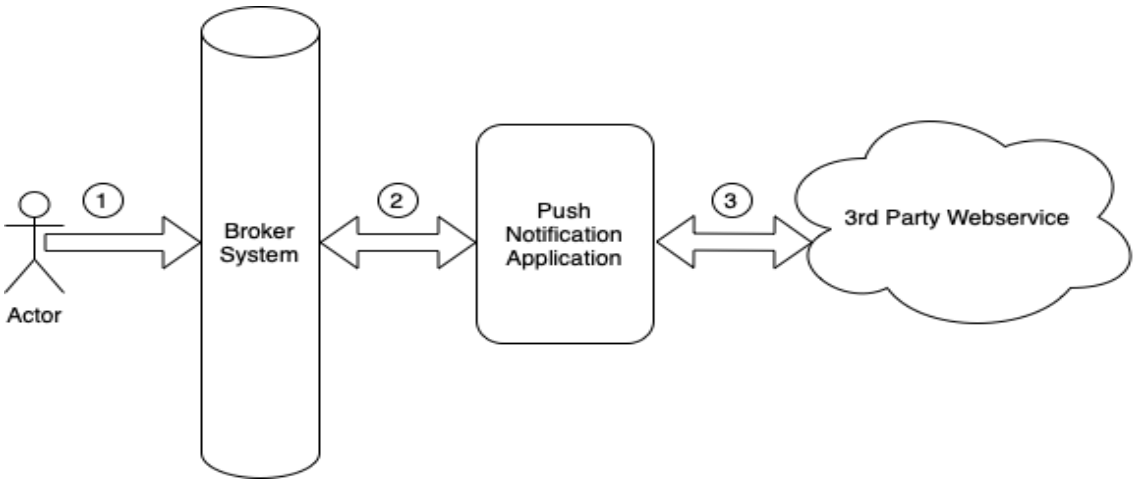
This document is written for developers who are implementing solutions using the NCBA Lipa Na M-Pesa Payment Notification service.

System Overview

When a payment is made through an NCBA Paybill/Buygoods/Till that has been marked to send 'API' Notification, the transaction details are queued for delivery by our application server to the client's/3rd party endpoint.

Upon sending the notification, the third party endpoint replies with appropriate response that is then logged in our systems.

This is the overall flow of the push notification service:



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Key:

1. A payment is received on the specific Paybill or Till
2. The system generates and shares the notification details with the push notification application
3. The push notification application sends JSON or Soap request/XML request to the 3rd party endpoint provided by the client.
4. The third party endpoint then sends aappropriate response depending on whether the notification passed the validation on the client's end.

Security Considerations

For 3rd Parties that opt to use our push service via the open web and have provided an http/ https endpoint the following details are required:

1. **UserName** to be used while pushing the notification to the 3rd Party endpoint.
2. **Password** to be used while pushing the notification to the 3rd Party endpoint.
3. **Secret Key** to be used while pushing the notification to the 3rd Party endpoint.

Notification Endpoint Spec Guide

Customers will be able to choose between receiving notifications either in JSON or XML format.

HTTP Request Specifications

The push notifications will be sent as a HTTP request to the provided endpoint. Below are the specifications to which the endpoint must abide:

- The endpoint should be capable of receiving data with either "application/xml" or "application/json" as the Content-Type, in accordance with the configuration specified on the portal, reflecting the payload format preferred by the customer/third party.
- The endpoint must accept POST requests.

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Paybill level Push Notifications may take the form of either Buygoods/Till Push Notifications, Paybill Push Notifications or NCBA Till Push Notifications and both can be presented in either XML or JSON format.

Parameter Value Descriptions Table

The table below outlines the meanings of each tag. It specifies the parameters for both JSON and XML payloads concerning Paybill Level Push Notifications for Buygoods and Paybill short codes.

Parameter	Description	Example
<TransType>	This is a String that represents the type of transaction that was done. May be TILLNUMBER or Pay Bill or PAYBILL	Pay Bill
<TransID>	This is a unique alphanumeric value that identifies the transaction i.e. M-Pesa reference	RKH7KICX13
<FTRef>	This is an alphanumeric value which is unique for each transaction. It refers to core banking reference number.	FTC231117BMWZ
<TransTime>	This is a numeric datetime value in the form of YYMMDDhhmmss	20231117100829
<TransAmount>	This is a positive numeric integer representing the transaction amount. May be one or two decimal places i.e. 1.0 or 1.00	1.0
<AccountNr>/ <BusinessShortCode>	This is is a numeric Paybill or Buygoods number that facilitates mobile payments.	880100
<BillRefNumber>/ <Narrative>	This is an alphanumeric string. For dedicated customer Paybill, this will be the narrative captured by sender while making the payment. This field may be empty or null or N/A. For NCBA Till, this field represents the short code allocated to the customer.	Rent-005 Or 000001
<Narrative>	This is an alphanumeric value that represents a variety of possible additional details quoted by a payer when initiating a transaction via NCBA Till. This field may be empty	Rent123
<PhoneNr>/<Mobile>	This is a numeric value that represents the phone number of the transaction. This value can also be a 64 String/varchar based on SHA-256 algorithm hashing on the phone number.	254711111111

<CustomerName>/<name>	This is an alphanumeric value that represents the name of the transaction initiator.	John Doe
<TransAmount>	This is a positive numeric integer representing the transaction amount.	1.0
<User>/<Username>	This is an alphanumeric string provided by the 3rd party for authentication purposes at the 3rd party's convenience.	Abc123!@#
<Password>	This is an alphanumeric string provided by the 3rd party for authentication purposes at the 3rd party's convenience.	Abc123!@#
<SecretKey>/<Hash>	This is an alphanumeric string that is generated by merging notification parameters, adding an extra value, hashing the combined data, and encoding the result.	YTAyMTEyZTNiN2U5MGQ1MGMyNjdmMzllY2FiOWFhMzYxZjM4ZDgyYTRjNDkzMWQ3NjlkYzY4NWMyNjE4MWEwZg==
<Status> (SUCCESS)	This is a fixed status string parameter in the XML payload that is assigned to each notification payload.	SUCCESS
<created_at>	This is a numeric datetime value in the form of YY-MM-DD-HH:mm:ss.S of when the notification was created.	2023-11-17 10:36:30.0

SecretKey Generation

The HashVal/Hash is generated by concatenating the secret key, notification details and '1' in a specific order, hashing with SHA256 and converting the hashed bytes into a Base64 string. You can use the below code samples as examples of how one can generate the hash or you can reverse engineer the algorithm in order to decode the hashval.

JAVA Hashing Version

```
public String hashPushRequest(String secretKey) {
    try {
        //secretKey -> This is an alphanumeric string provided by the 3rd party for
        authentication purposes at the 3rd party's convenience.

        //Concatenate values together to form a string
        String hashString = secretKey + TransType +TransID + TransactionTime + TransAmount +
        CreditAccount + BillRefNumber + Mobile + Name + "1";

        String sha256hex =
```


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```
        Hashing.sha256().hashString(hashString,StandardCharsets.UTF_8).toString();

        byte[] bytesEncodedHash = Base64.encodeBase64(sha256hex.getBytes());

        return new String(bytesEncodedHash);

      } catch (Exception e) {
        return null;
      }
    }
```

1. Buygoods Push Notifications

a) JSON Body Specifications

This refers to a scenario where a third-party endpoint is set up to receive Push Notifications in JSON format. Below is an example of the JSON payload to be transmitted along with the anticipated response.

JSON Request to be sent


```
{
        "TransType": "TILLNUMBER",
        "TransID": "RKH71L7YCD",
        "TransTime": "20231117103607",
        "TransAmount": "10.00",
        "BusinessShortCode": "488488",
        "BillRefNumber": "N/A",
        "Mobile": "254711111111",
        "name": "JOHN DOE",
        "created_at": "2023-11-17 10:36:30.0",
        "Username": "Abc123!@#",
        "Password": "Abc123!@#",
        "Hash":
        "ZDBjNzNjZTc5MGMwZjBhMjhmYjYxNDQyZTlkNDM4YjhlNzUxZjliMzUyOTYxMGFlYzNhN2NkZDBkOWMwYmNjZA\u003d\u003d
        "
}
```


JSON Response Specifications

The response to the JSON body sent to the 3rd party endpoint must also be in the JSON form. Below are samples:

Success Response


```
{
  "ResultCode": "0",
  "ResultDesc": "Test API Response"
}
```


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Failed Response


```
{
  "ResultCode": "1",
  "ResultDesc": "Failed Test API Response"
}
```


The above represents a successful JSON response and unsuccessful json response. The push notification service reads the values passed and logs it.

b) XML Body Specifications

This refers to a scenario where a third-party endpoint is set up to receive Push Notifications in XML format. Below is an example of the XML payload to be transmitted along with the anticipated response.

XML Request to be sent


```
<soapenv:Envelope
  xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Header/>
  <soapenv:Body>
    <NCBAPaymentNotificationRequest>
          <User> Abc123!@#</User>
                <Password> Abc123!@#</Password>
                <HashVal>NjkwOWEzYTY0YTFmMTFhMmYwMTUwYmU1NTNiYTg5NmEwYTkxNzlhZjI4YjYyOTFjMmZjYjhjNjdlOGQ1O
                WQ2ZA==</HashVal>
          <TransType>TILLNUMBER</TransType>
          <TransID>RK82R1QUCK</TransID>
          <TransTime>20231108142416</TransTime>
          <TransAmount>10.00</TransAmount>
          <AccountNr>488488</AccountNr>
          <Narrative>N/A</Narrative>
          <PhoneNr>254711111111</PhoneNr>
          <CustomerName>JOHN DOE</CustomerName>
          <Status>SUCCESS</Status>
    </NCBAPaymentNotificationRequest>
  </soapenv:Body>
</soapenv:Envelope>
```


XML Response Specifications

The response to the XML body sent to the 3rd party endpoint must also be in the XML form. Below are samples:

Success Response

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Header/>
  <soapenv:Body>
    <NCBAPaymentNotificationResult>
      <Result>OK</Result>
    </NCBAPaymentNotificationResult>
  </soapenv:Body>
</soapenv:Envelope>
```

Failed Response

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Header/>
  <soapenv:Body>
    <NCBAPaymentNotificationResult>
      <Result>FAIL</Result>
    </NCBAPaymentNotificationResult>
  </soapenv:Body>
</soapenv:Envelope>
```

The above represents a successful xml response and unsuccessful xml response. The push notification service reads the value passed in the <Result> tag and and logs it.

2. Paybill Push Notification

a) JSON Body Specifications

In this scenario, the third-party endpoint is configured to receive Push Notifications in JSON format for a specific Paybill short code. Below is an example of the JSON payload to be transmitted along with the anticipated response.

JSON Request to be sent

```
{
        "TransType": "PAYBILL",
        "TransID": "RK70O1A9HU",
        "TransTime": "20231107163508",
        "TransAmount": "100.0",
        "BusinessShortCode": "488488",
        "BillRefNumber": "Rent-005",
        "Mobile": "254711111111",
        "name": "JOHN DOE",
        "created_at": "2023-11-07 16:35:37.0",
        "Username": "Abc123!@#",
```

```
    "Password": "Abc123!@#",
    "Hash":
    "NmUxNmNkN2NiMGI1Y2IzMmNhMjdmMGI5ZDRkZDljNTRhMDAxZjg2MDE1ZDgzYWRkNjM2YzlhZTcxZTNmNjcwNg\u003d\u003d"
}
```


JSON Response Specifications

The response to the JSON body sent to the 3rd party endpoint must also be in the JSON form. Below are samples:

Success Response


```
{
  "ResultCode": "0",
  "ResultDesc": "Test API Response"
}
```


Failed Response


```
{
  "ResultCode": "1",
  "ResultDesc": "Failed Test API Response"
}
```


The above represents a successful json response and unsuccessful json response. The push notification service reads the values passed and logs it.

b) XML Body Specifications

In this scenario, the third-party endpoint is configured to receive Push Notifications in XML format for a specific Paybill short code. Below is an example of the XML payload to be transmitted along with the anticipated response.

XML Request to be sent


```
<soapenv:Envelope
  xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Header/>
  <soapenv:Body>
    <NCBAPaymentNotificationRequest>
        <User>Abc123!@#</User>
            <Password>Abc123!@#</Password>
            <HashVal>YmNkM2FiYzU1ZGY4N2RjMzRmZjZkNTBiMTgzYTIxZWQ0NmQxYTRkOTM2ZTE5ZTQ1ODI2OGIwMzQ3MTdjY
            TdiYg==</HashVal>
        <TransType>Pay Bill</TransType>
```


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```
            <TransID>RK88R1NJ58</TransID>
            <TransTime>20231108141412</TransTime>
            <TransAmount>100.00</TransAmount>
            <AccountNr>488488</AccountNr>
            <Narrative>Rent-005</Narrative>
            <PhoneNr>254711111111</PhoneNr>
            <CustomerName>JOHN DOE</CustomerName>
            <Status>SUCCESS</Status>
        </NCBAPaymentNotificationRequest>
    </soapenv:Body>
</soapenv:Envelope>
```

XML Response Specifications

The response to the XML body sent to the 3rd party endpoint must also be in the XML form. Below are samples:

Success Response

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
   <soapenv:Header/>
   <soapenv:Body>
      <NCBAPaymentNotificationResult>
         <Result>OK</Result>
      </NCBAPaymentNotificationResult>
   </soapenv:Body>
</soapenv:Envelope>
```

Failed Response

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
   <soapenv:Header/>
   <soapenv:Body>
      <NCBAPaymentNotificationResult>
         <Result>FAIL</Result>
      </NCBAPaymentNotificationResult>
   </soapenv:Body>
</soapenv:Envelope>
```

The above represents a successful xml response and unsuccessful xml response. The push notification service reads the value passed in the <Result> tag and and logs it.

3. NCBA Till (Alias) Push Notification under Paybill No. 880100

a) JSON Body Specifications

This is where a third-party endpoint is set up to receive NCBA Till Push Notifications in JSON format. Below is an example of the JSON payload to be transmitted along with the anticipated response.

JSON Request to be sent


```
{
        "TransType": "Pay Bill",
        "TransID": "RK91U5J2AD",
        "FTRef": "FTC231009BMZG",
        "TransTime": "20231109112200",
        "TransAmount": "2.0",
        "BusinessShortCode": "880100",
        "BillRefNumber": "000001",
        "Narrative": "Rent123",
        "Mobile": "254711111111",
        "name": "JOHN DOE",
        "Username": "Abc123!@#",
        "Password": "Abc123!@#",
        "Hash":
        "Y2Y3NWUzMjJkZTVhMmQ3NzYzNzU1NmZiYjk1MTkyNmU0NmZlNzA1ZmU0NjY0NGFhZmQyOTA5M2NiOWU3YmIwMQ\u003d
        \u003d"
}
```


JSON Response Specifications

The response to the JSON body sent to the 3rd party endpoint must also be in the JSON form.

Below are samples:

Success Response


```
{
  "ResultCode": "0",
  "ResultDesc": "Test API Response"
}
```


Failed Response


```
{
  "ResultCode": "1",
  "ResultDesc": "Failed Test API Response"
}
```


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b) XML Body Specifications

In the scenario where the third party is set up to receive XML notifications, provided below is an example of the XML payload to be transmitted along with the anticipated response.

XML Request to be sent

```
<soap:Envelope xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xmlns:xsd="http://www.w3.org/2001/XMLSchema"
        xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <CBAPaymentRequest xmlns="https://live.mzizi.co.ke/cbapayment">
          <Mobile>254711111111</Mobile>
          <FTRef>FTC240111ZOGI</FTRef>
          <BillRefNumber>000001</BillRefNumber>
          <Narrative>Rent123</Narrative>
          <BusinessShortCode>880100</BusinessShortCode>
          <TransType>Pay Bill</TransType>
          <TransTime>20240111100431</TransTime>
          <TransAmount>2.0</TransAmount>
          <Name>JOHN DOE</Name>
          <TransID>SAB2OVHZDK</TransID>
          <Username>Abc123!@#</Username>
          <Password>Abc123!@#</Password>
                <Status>SUCCESS</Status>
                <SecretKey>ZGU2MzE5Nzc0ZTc4NDA3ODA4NjA1YWJhZTE3YjBmMjhlODNiNmIyMTZkOWJiNjNmYWM1Mzk5NjhlOGIx
                NmJmZQ==</SecretKey>
    </CBAPaymentRequest>
  </soap:Body>
</soap:Envelope>
```

XML Response Specifications

The response to the XML body sent to the 3rd party endpoint must also be in the XML form. Below are samples:

Success Response

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Header/>
  <soapenv:Body>
    <NCBAPaymentNotificationResult>
      <Result>OK</Result>
```


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```
    </NCBAPaymentNotificationResult>
  </soapenv:Body>
</soapenv:Envelope>
```

Failed Response

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Header/>
  <soapenv:Body>
    <NCBAPaymentNotificationResult>
      <Result>FAIL</Result>
    </NCBAPaymentNotificationResult>
  </soapenv:Body>
</soapenv:Envelope>
```


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